

Substations		Technical Procedure	
Equipment Insulation Capacitance and Tan δ Dissipation Factor for Two Winding Transformer type CB 100 TEOF		Document No: T-025-r0	
		Issued to: Networks	
		Status: APPROVED	
Procedure: Testing Procedure(s)		Approved Date: 05 Feb 2001	
		Date to be Reviewed: Feb 2006	

Introduction

This document lists the equipment Testing Procedure checklist, which must be performed for Power Transformers. According to each manufacturer Instructions.

Safety Precautions

- Isolate the transformer " out of service "
- A word order permit must be issued.
- The qualified test grup staff should carry the suitable safety category
- Safety fence (rope) with caution marks surrounding the work area,
- Disconnect the transformer from bridles
- Connect the bridles to the earth.
- All bushings should be clean other wise errors reading could be happened
- Make short circuit SC for HV side and LV side bushings
- Connect the earth of the transformer to the earth point of the tester

Tools and Equipment

- Test equipment type CB100-TEOF
AVO - International, 4651 Single Westmorland TX 75237 - 1017 USA

Work to be Carried Out

TEST # 1 (C_{HL} + C_H)

1. Test mode " GST L - Ground "
2. Connect HV Cable to S.C. HV Bushing
3. Connect LV cable to S.C. LV bushing

4. Turn-on to C position and dial switches of the tester to get balance of pointer.
5. Changeover the switch to " DF " Positions .And select the range and dial to get balance of pointer.
6. Record the results on the attached test sheet
7. Correct the reading to be at 20 °C.

Test # 2 C_H

1. Test mode " GST 1 - Guard " Keep all pervious connection.
2. Repeat steps 4,5,7 On the pervious test.
3. Record the results on the attached test sheet

Test # 3 C_{HL}

1. Test mode " UST " Keep all pervious connection.
2. Repeat steps 4, 5, 7 on the test #1.
3. Record the results on the attached test sheet

Test # 4 $C_L + C_{HL}$

1. Test mode " GSTL- ground " . Connect HV cable of the to S.C.. LV bushing
2. Connect LV cable of the to S.C.. HV bushing.
3. Repeat steps 4, 5, 7 on the test #1.
4. Record the results on the attached test sheet

Test # 5 C_H

1. Test mode " GST H-guard " keep all pervious connections.
2. Repeat steps 4, 5, 7 on the test #1.
3. Record the results on the attached test sheet.

Test # 6 C_{HL}

1. Test mode " UST-ground " keep all pervious connections.
2. Repeat steps 4, 5, 7 on the test #1.
3. Record the results on the attached test sheet.

Test Sheet

High Voltage Tan δ testing (Results corrected to be at 20 °C)

Test #	Insulation tested		Tan δ %	Pfd Cap.	Notes
1	CH + CHL				
2	CH				
3	CHL (UST)				
4	CHL				
5	CL + CHL				
6	CL				
7	CHL (UST)				
8	CLH				

High Voltage Tan δ testing (Results corrected to be at 20 °C)

Test #	Insulation tested		Tan δ %	Pfd Cap.	Notes
1	CH + CHL				
2	CH				
3	CHL (UST)				
4	CHL				
5	CL + CHL				
6	CL				
7	CHL (UST)				
8	CLH				
9	CT + CHT				
10	CT				
11	CHT (UST)				
12	CLT				

Location:
Checked by:

Equipment code:
Date:

Sig